

Q.1) FULL OUTER JOIN.

Query:

```
SELECT *  
FROM CLASSROOM  
LEFT JOIN SUBJECTT  
ON CLASSROOM.SUB_ID = SUBJECTT.SUB_ID  
UNION  
SELECT*  
FROM CLASSROOM  
RIGHT JOIN SUBJECTT  
ON CLASSROOM.SUB_ID= SUBJECTT.SUB_ID;
```

Screenshot:

The screenshot shows the MySQL Workbench interface. The query editor contains the following SQL query:

```
SELECT *  
FROM CLASSROOM  
LEFT JOIN SUBJECTT  
ON CLASSROOM.SUB_ID = SUBJECTT.SUB_ID  
UNION  
SELECT*  
FROM CLASSROOM  
RIGHT JOIN SUBJECTT  
ON CLASSROOM.SUB_ID= SUBJECTT.SUB_ID;
```

The Results tab displays the output of the query, which is a table with 10 columns: CLASS_ID, ROOM_NAME, SUB_ID, CAPACITY, HAS_PROJECTOR, SUB_ID, SUB_NAME, DEPT_ID, TEACHER_ID, and CREDITS. The table contains 10 rows of data, including courses like 'Introduction to Database', 'Artificial Intelligence', 'Linear Algebra', 'Quantum Mechanics', and 'Ancient Civilizations'.

CLASS_ID	ROOM_NAME	SUB_ID	CAPACITY	HAS_PROJECTOR	SUB_ID	SUB_NAME	DEPT_ID	TEACHER_ID	CREDITS
C1	101-A	CS101	50	1	CS101	Introduction to Database	D1	T101	4
C2	102-B	CS102	35	1	CS102	Artificial Intelligence	D1	T102	4
C3	201	MA201	40	0	MA201	Linear Algebra	D2	T103	3
C4	Lab-3	PH301	25	1	PH301	Quantum Mechanics	D3	T104	4
C5	404	HI101	60	0	HI101	Ancient Civilizations	D4	T105	2

The bottom of the screenshot shows the 'Action Output' tab, which displays the execution details of the query, including the time taken and the number of rows affected.

Insights: This Query returns all columns from student and classroom table, with common and uncommon columns.